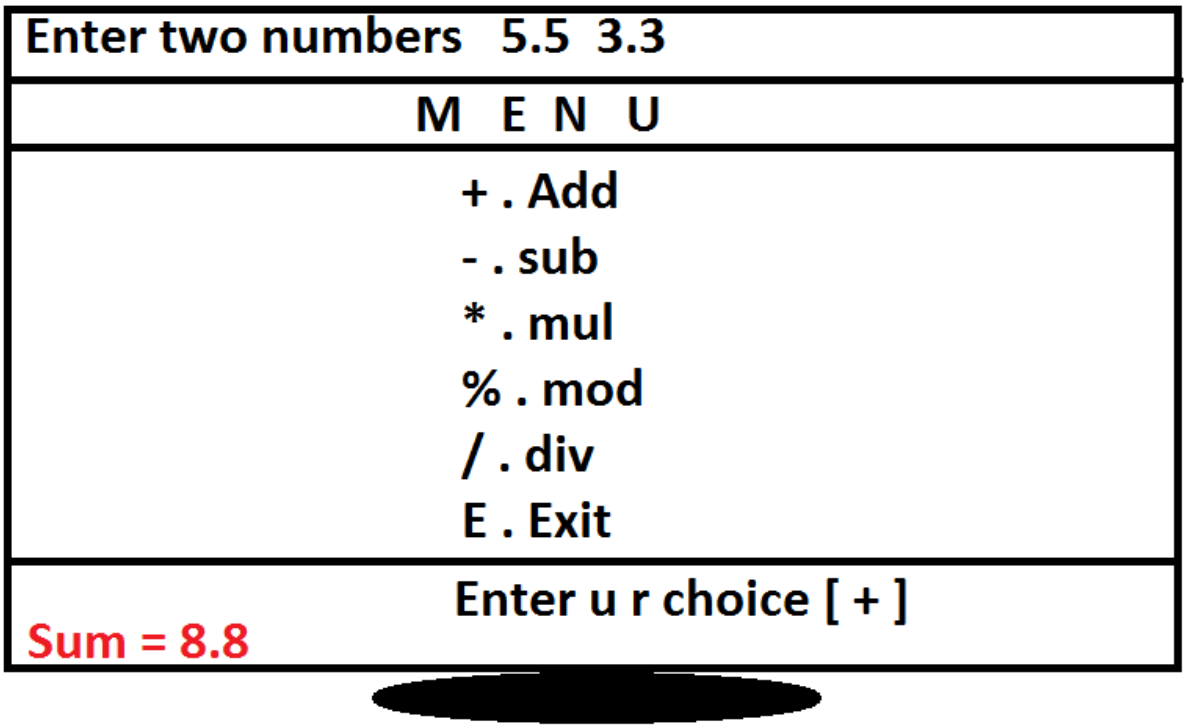
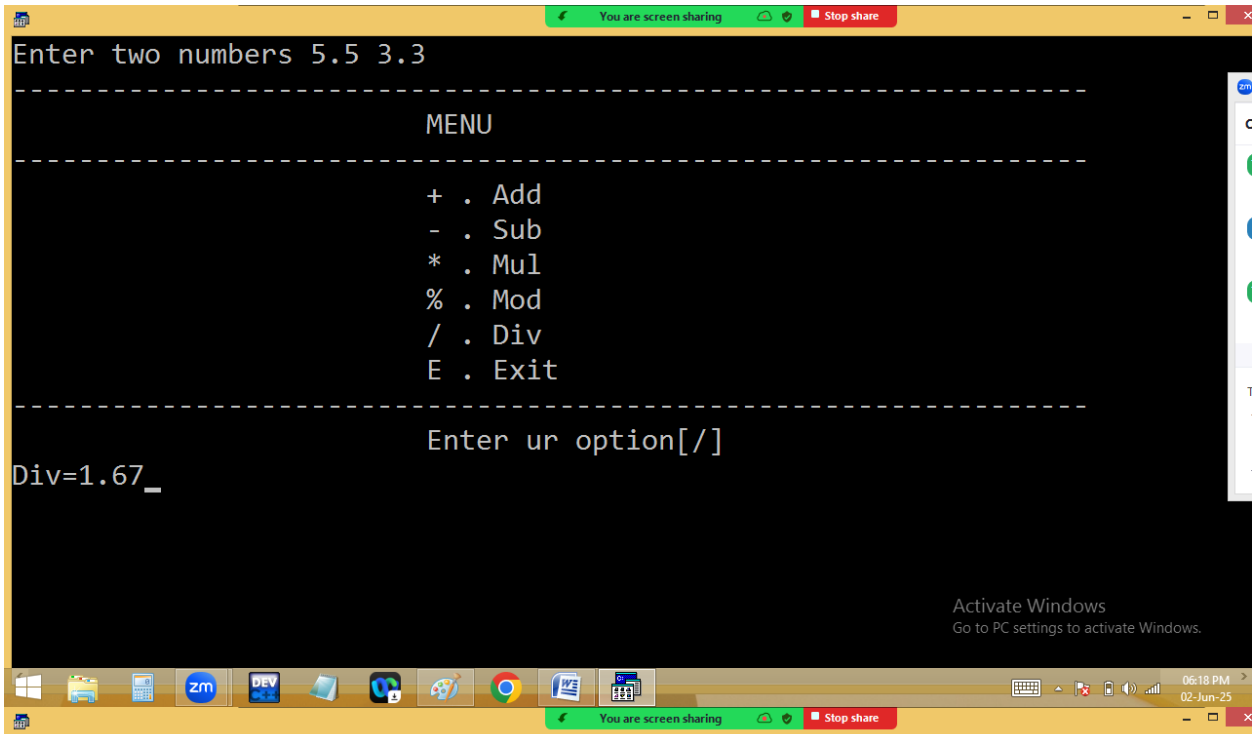
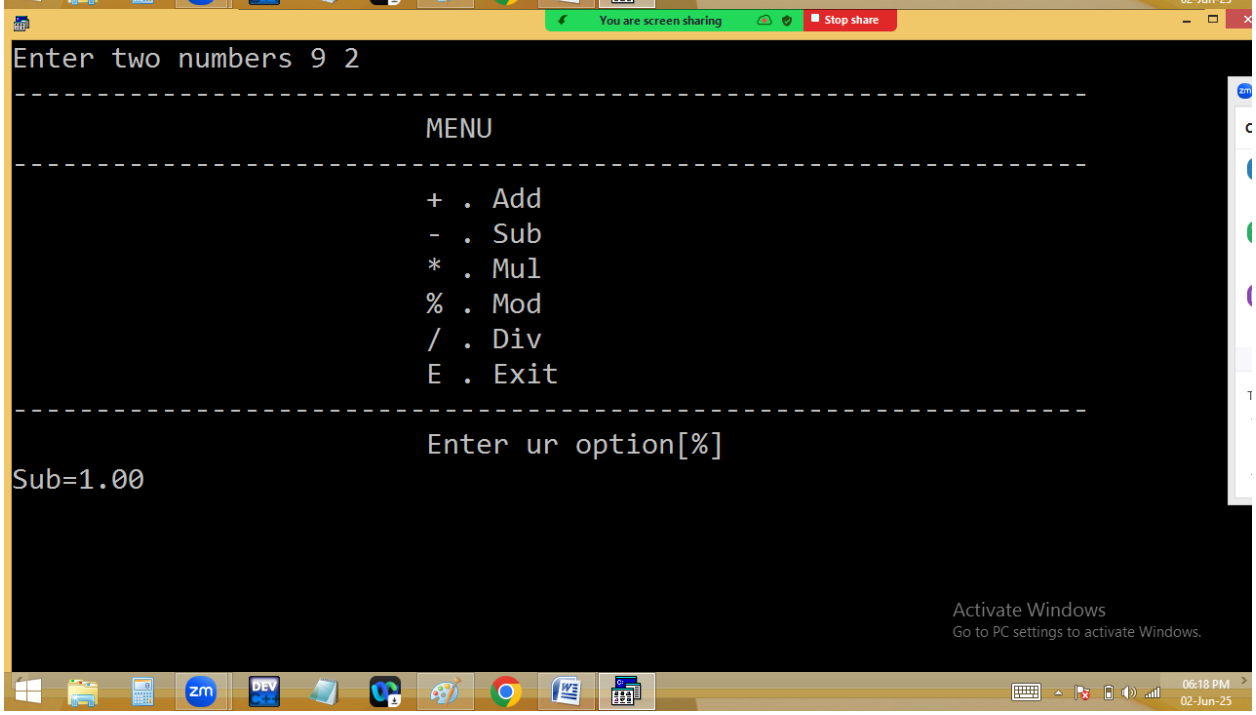
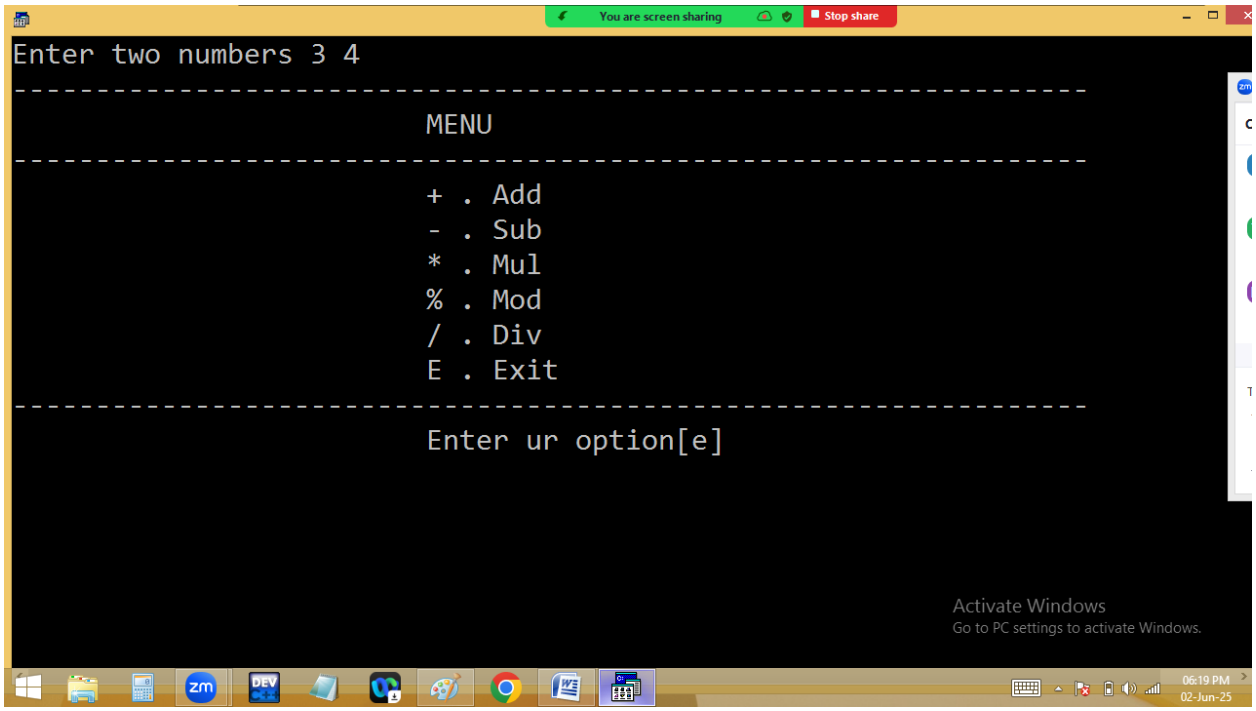
Read two numbers and perform arithmetic operation based on the user selection using switch?




You are screen sharing
Stop share
Enter two numbers 5.5 3.3
MENU
+ . Add
- . Sub
* . Mul
% . Mod
/ . Div
E . Exit
Enter ur option[/]
Div=1.67
Activate Windows
Go to PC settings to activate Windows.
06:18 PM
02-Jun-25


You are screen sharing
Stop share
Enter two numbers 9 2
MENU
+ . Add
- . Sub
* . Mul
% . Mod
/ . Div
E . Exit
Enter ur option[%]
Sub=1.00
Activate Windows
Go to PC settings to activate Windows.
06:18 PM
02-Jun-25



Loops / Iterations / Repetitive statements

Loops are used to repeat a block/group of statements continuously until the given condition becomes false.

Loops reduce program size and improves performance.

In loops beginning and ending points are same.

C-Language supports basically 2 types of loops.

1. **Entry/pre controlled loops.**
2. **Exit/post controlled loops.**

In entry control loops, condition is tested first and it is true then only statements block is executed.

Under entry control loops we are having

i. **While loop**
ii. **For loop**

In exit control loop, the statements are executed first and later condition is tested.

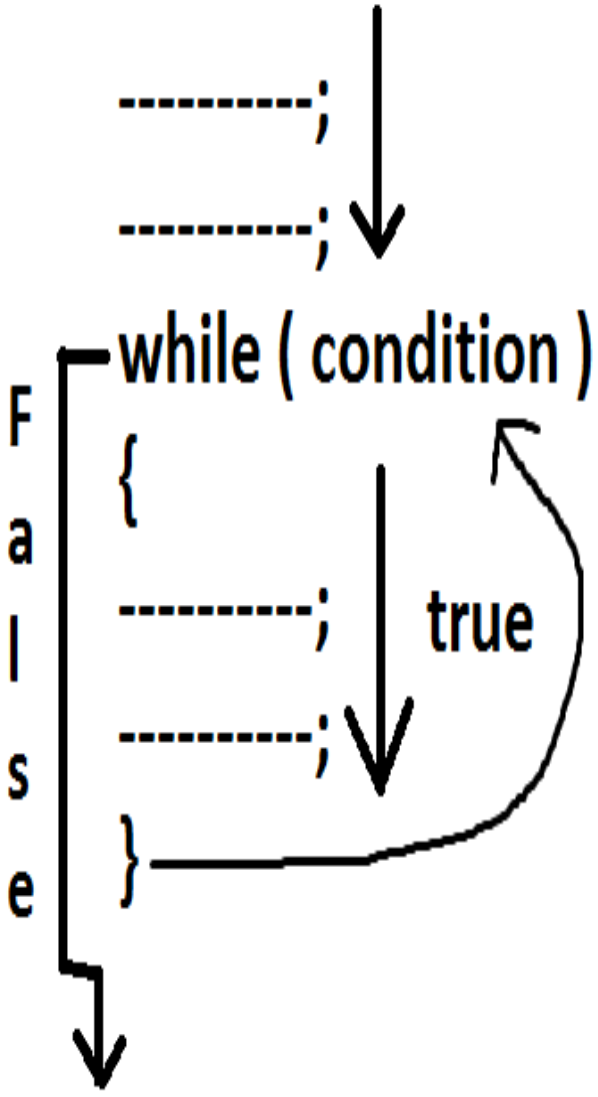
Under exit control loop we are having

i. **do while.**

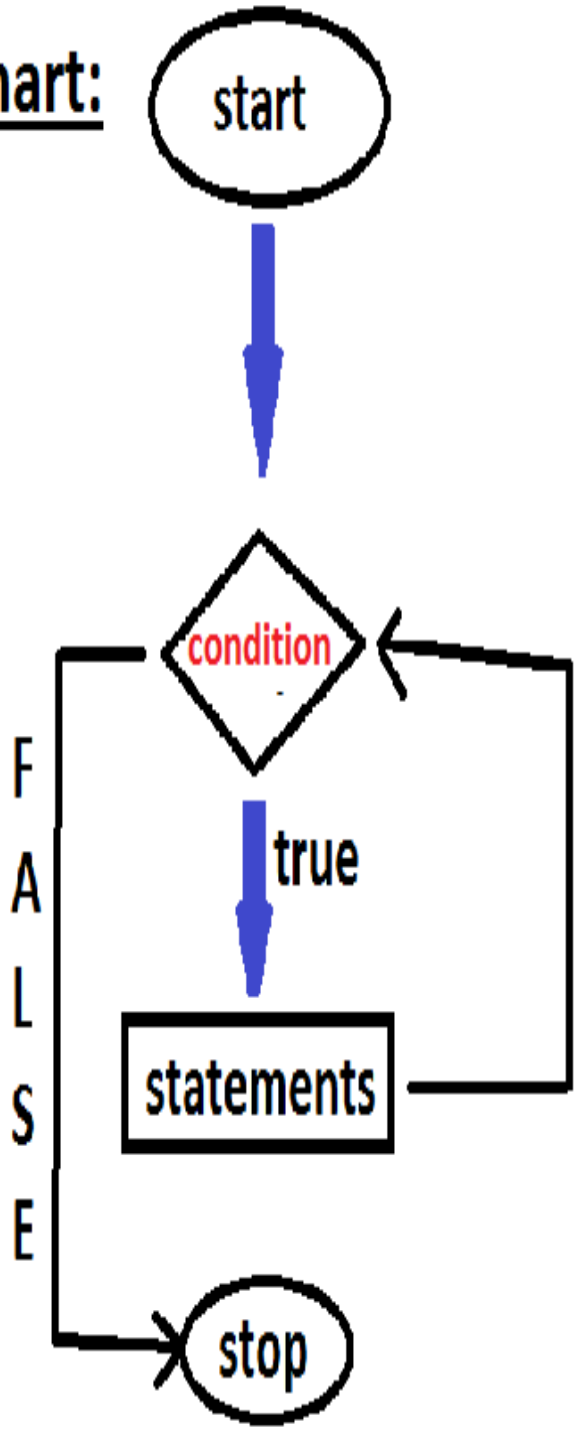
While loop:

- **while is a keyword.**
- **In while loop condition is tested first and it is true then only while block statements are executed. After executing while block statements, the program execution automatically shifted/jumped to while condition at the beginning. If it is true then once again the while block statements are repeated. Like this the process is continued until while condition becomes false.**
- **While is entry control loop.**

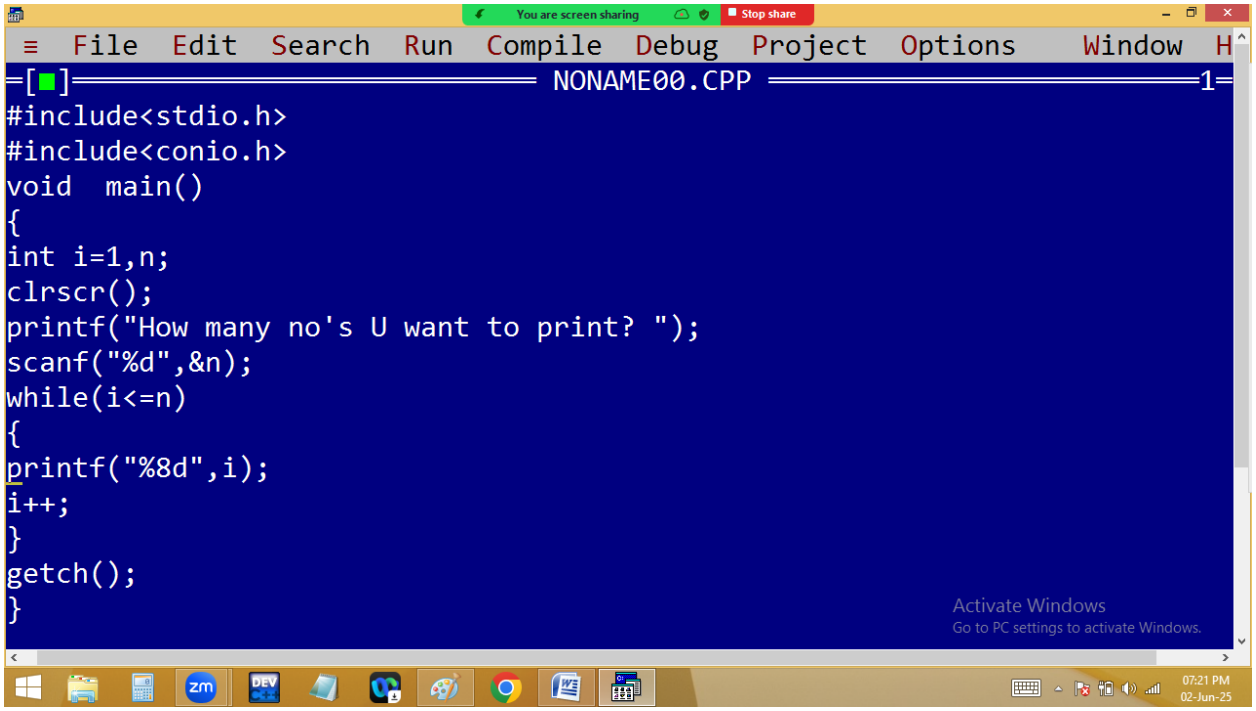
Syntax:

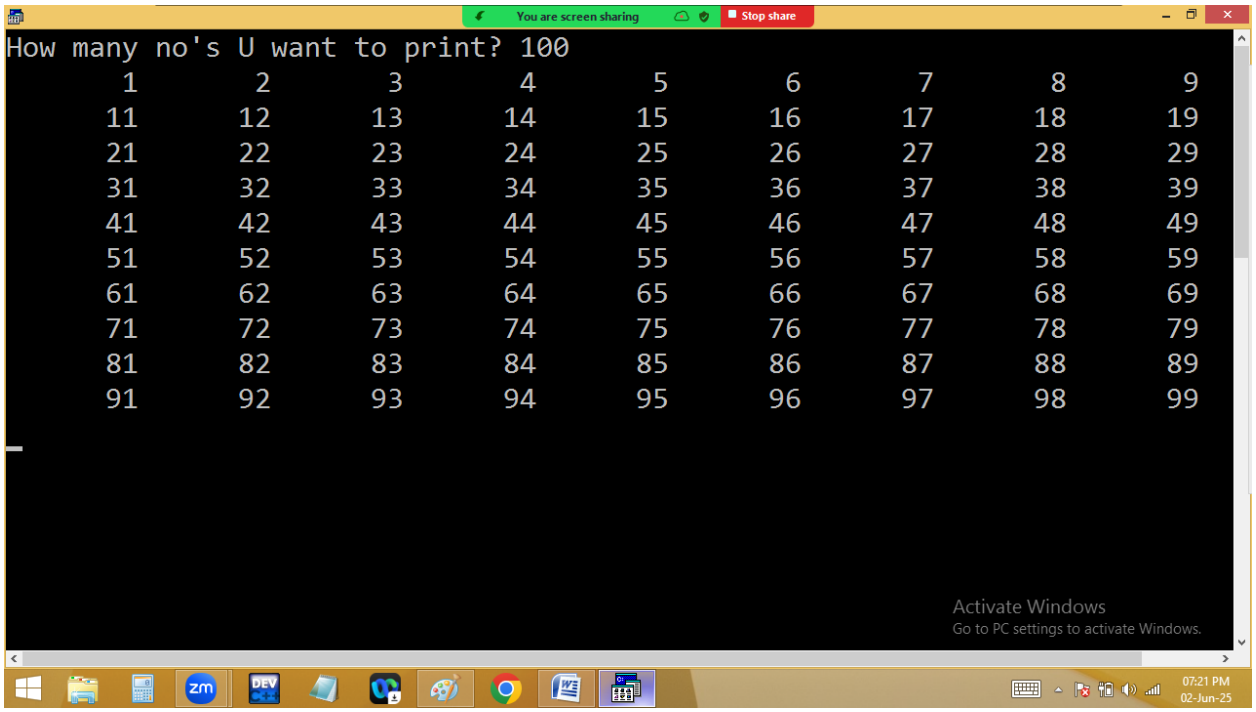


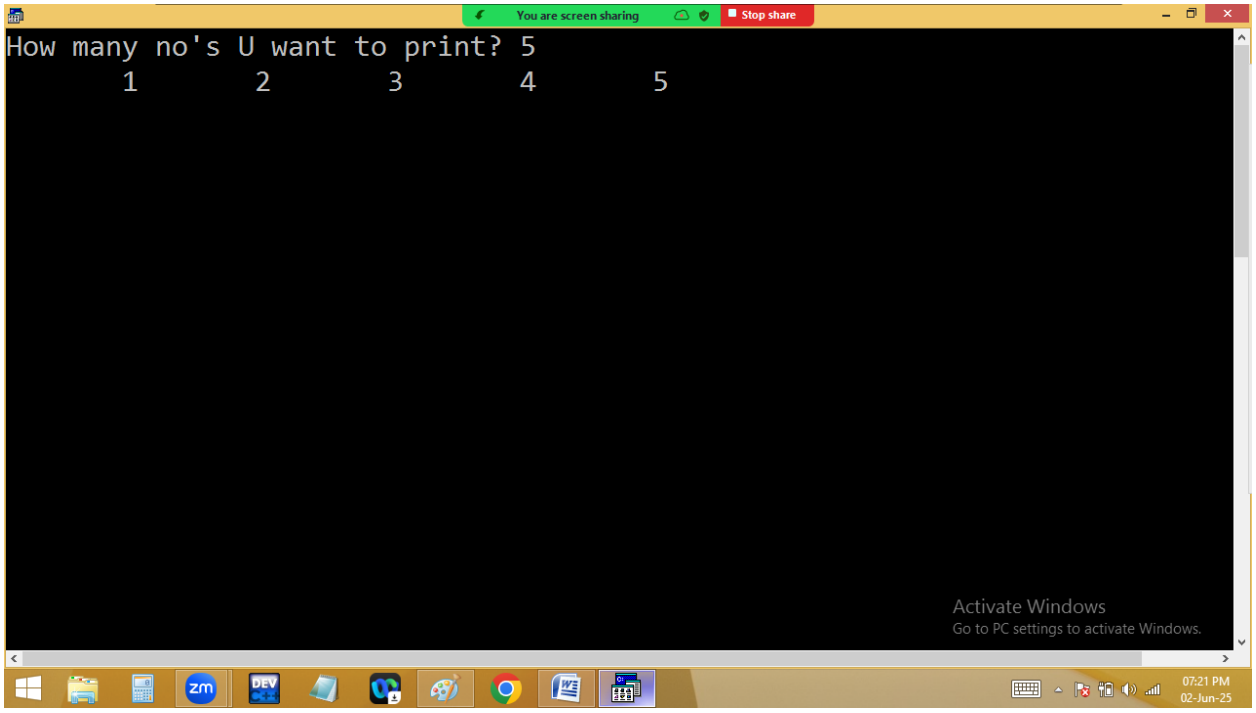
Flow chart:

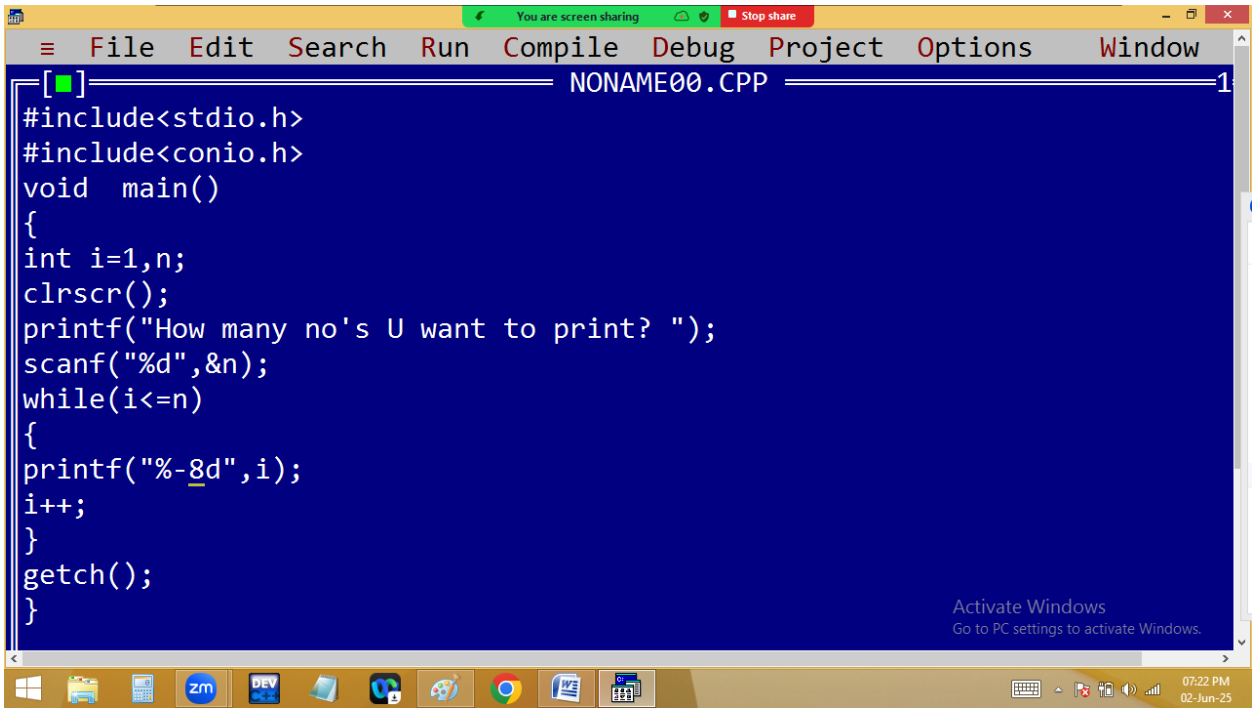


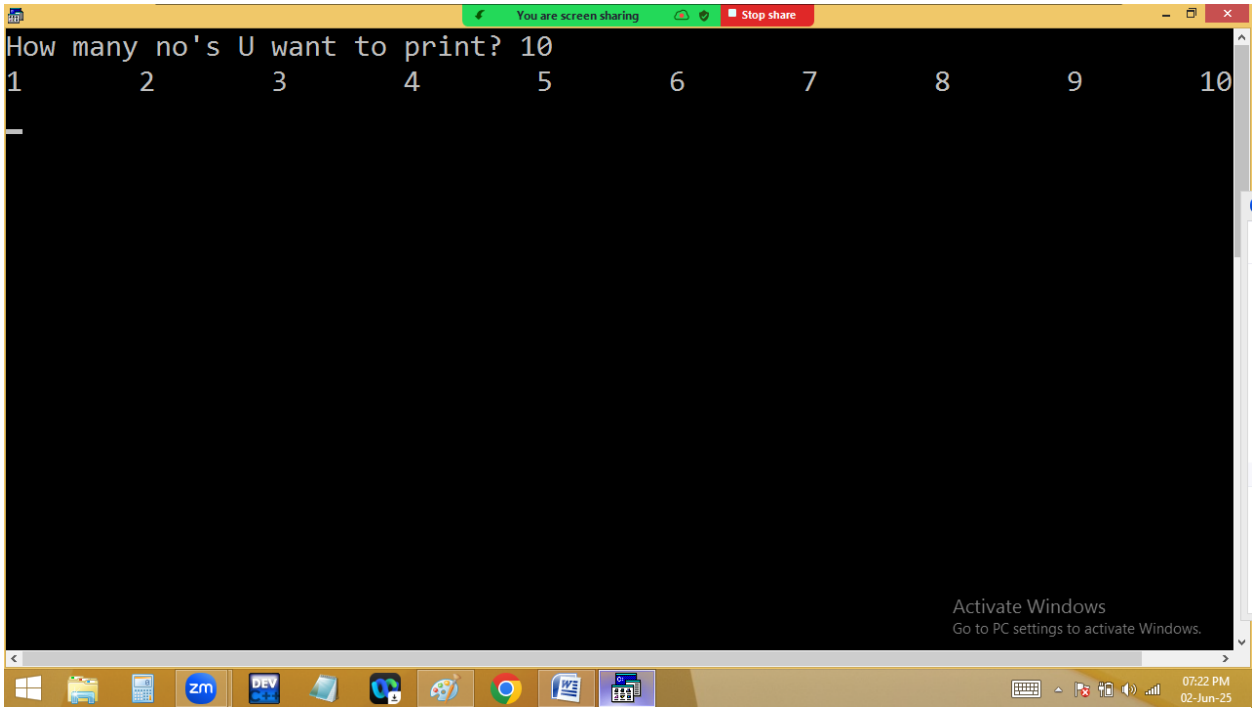
Eg: printing 1..n no’s using while?

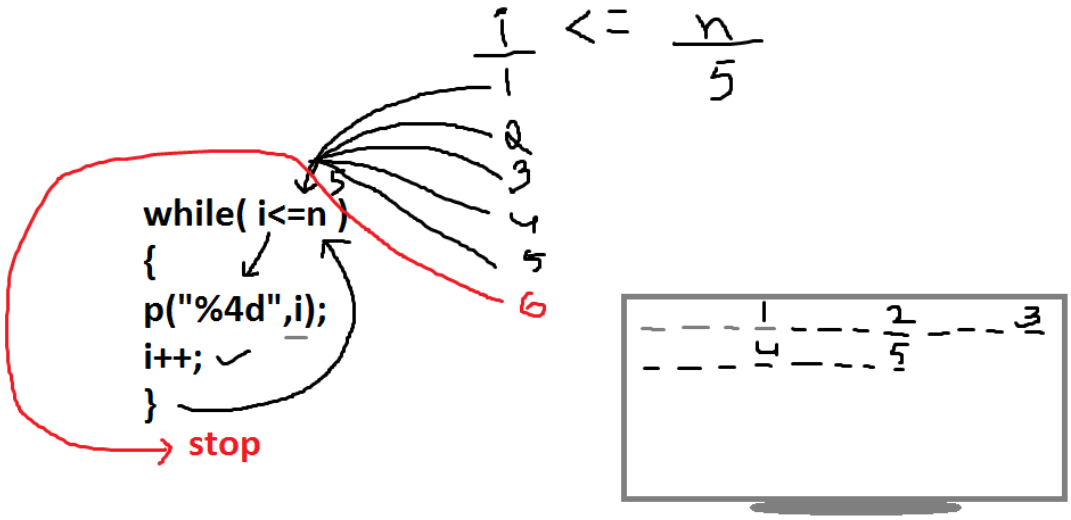




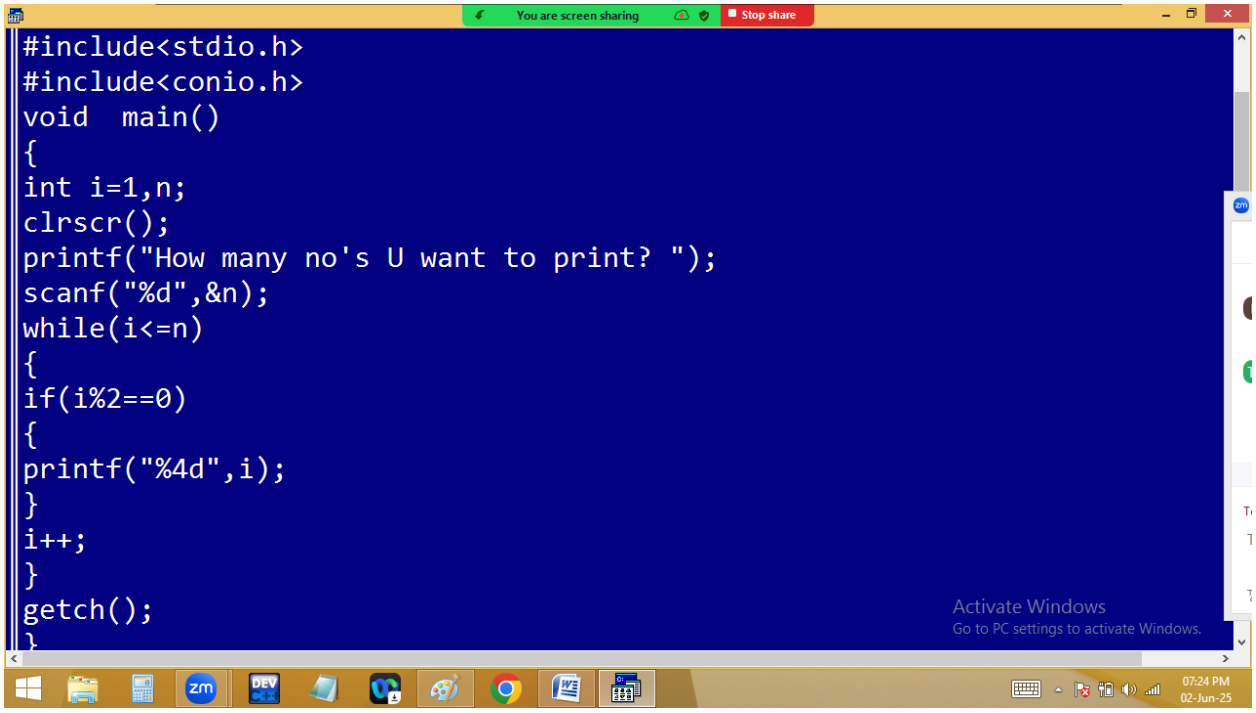

You are screen sharing
Stop share
How many no's U want to print? 5
1 2 3 4 5
Activate Windows
Go to PC settings to activate Windows.
07:21 PM
02-Jun-25

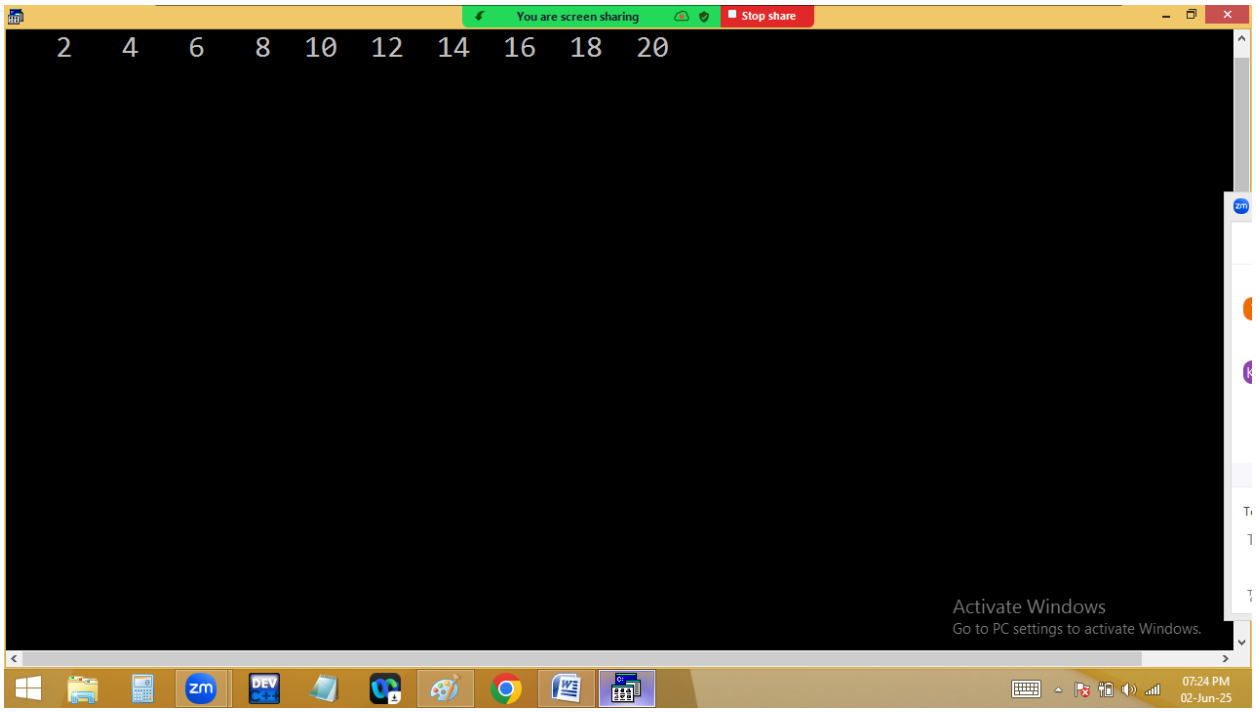

You are screen sharing
Stop share
File Edit Search Run Compile Debug Project Options Window
NONAME00.CPP
#include<stdio.h>
#include<conio.h>
void main()
{
int i=1,n;
clrscr();
printf("How many no's U want to print? ");
scanf("%d",&n);
while(i<=n)
{
printf("%-8d",i);
i++;
}
getch();
}
Activate Windows
Go to PC settings to activate Windows.
07:22 PM
02-Jun-25





Print 1..n even no's only?


You are screen sharing
Stop share
#include<stdio.h>
#include<conio.h>
void main()
{
int i=1,n;
clrscr();
printf("How many no's U want to print? ");
scanf("%d",&n);
while(i<=n)
{
if(i%2==0)
{
printf("%4d",i);
}
i++;
}
getch();
}
Activate Windows
Go to PC settings to activate Windows.
07:24 PM
02-Jun-25


You are screen sharing
Stop share
2 4 6 8 10 12 14 16 18 20
Activate Windows
Go to PC settings to activate Windows.
07:24 PM
02-Jun-25

Odd no’s:

```c
#include<stdio.h>
#include<conio.h>
void  main()
{
int i=1,n;
clrscr();
printf("How many no's U want to print? ");
scanf("%d",&n);
while(i<=n)
{
if(i%2!=0)
{
printf("%4d",i);
}
i++;
}
getch();
}
```

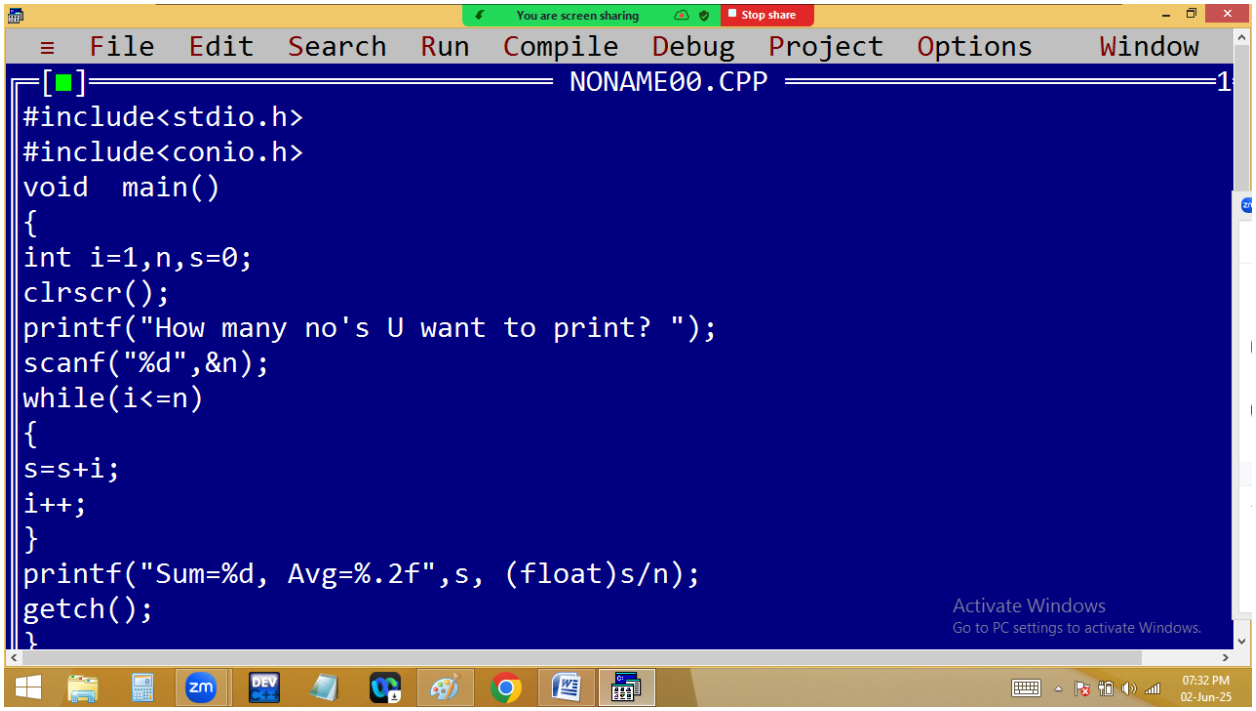
You are screen sharing Stop share
Activate Windows
Go to PC settings to activate Windows.
07:25 PM
02-Jun-25

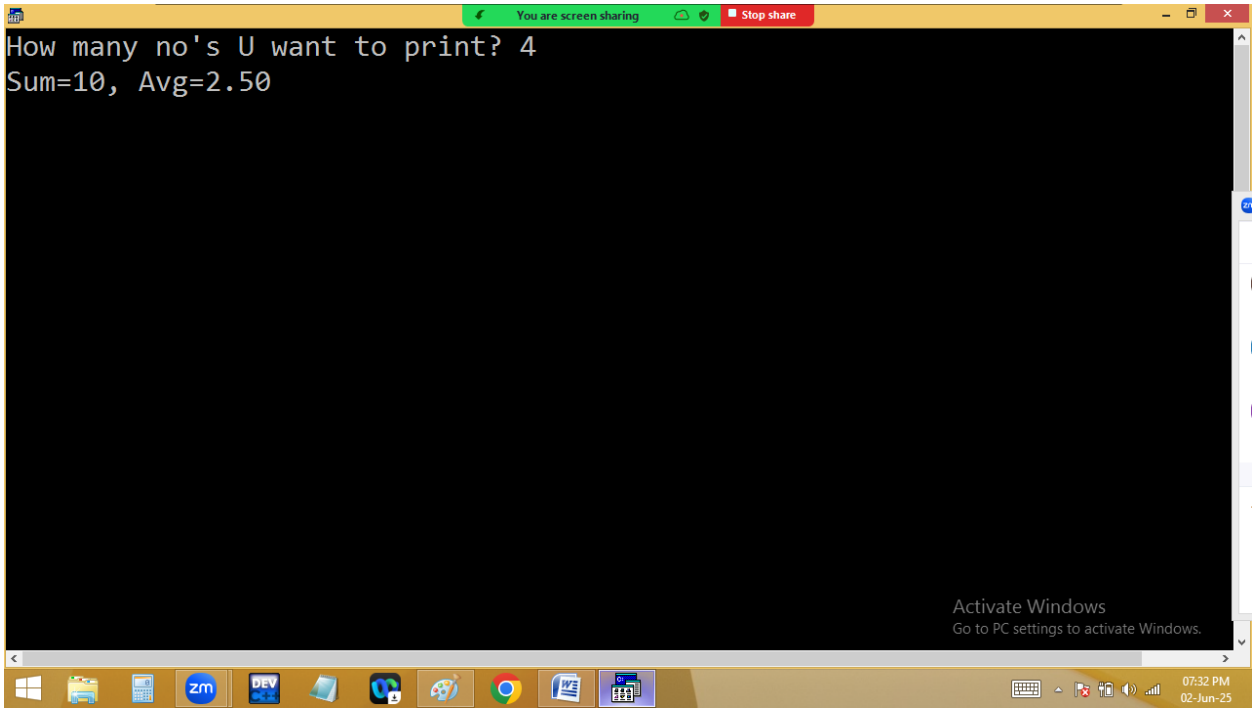
```
How many no's U want to print? 10
   1   3   5   7   9
```

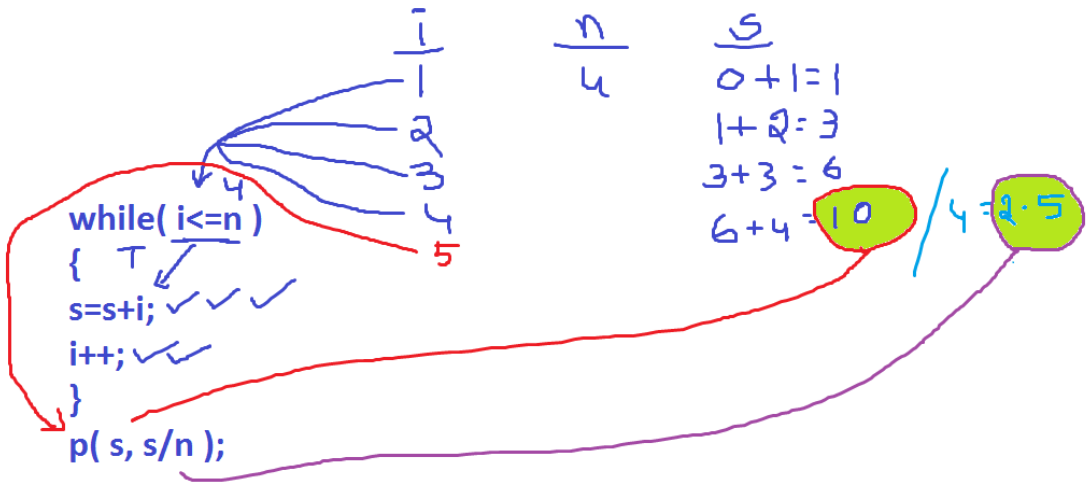
You are screen sharing Stop share
Activate Windows
Go to PC settings to activate Windows.
07:25 PM
02-Jun-25

Find 1..n no’s sum and mean [avg]:

Eg: n=4→ 1+2+3+4=10 ← sum, 10/4=2.5← avg



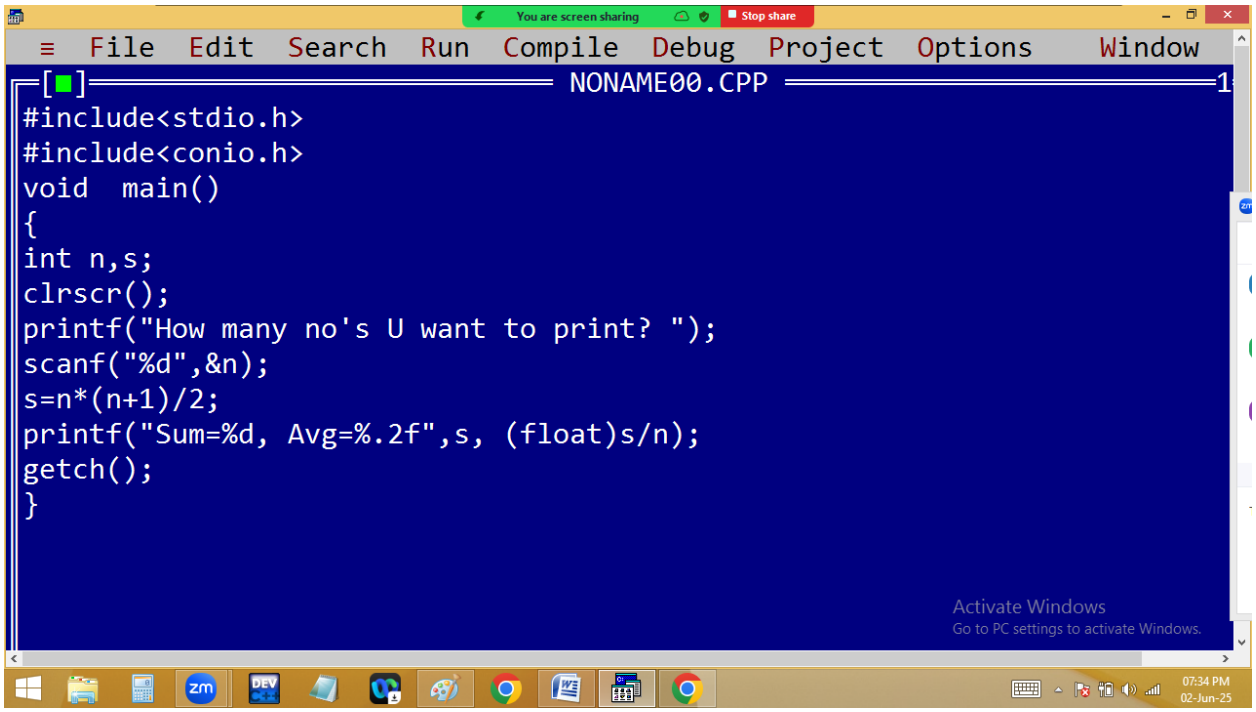


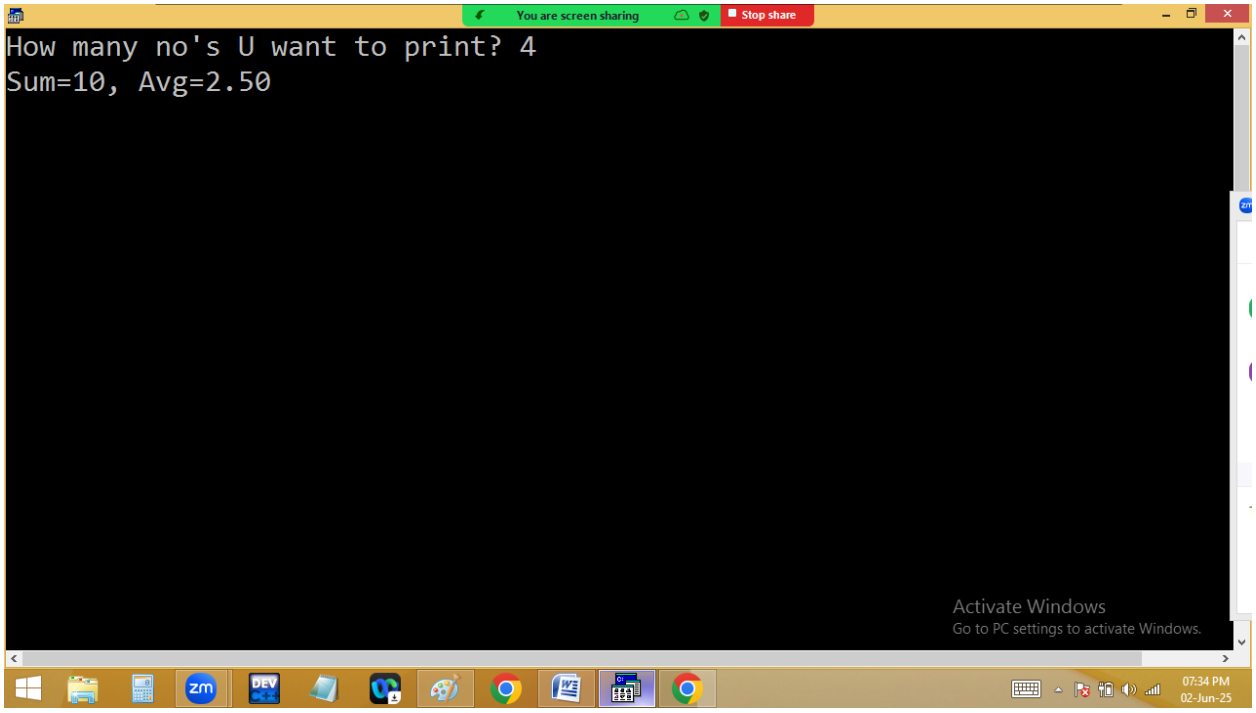


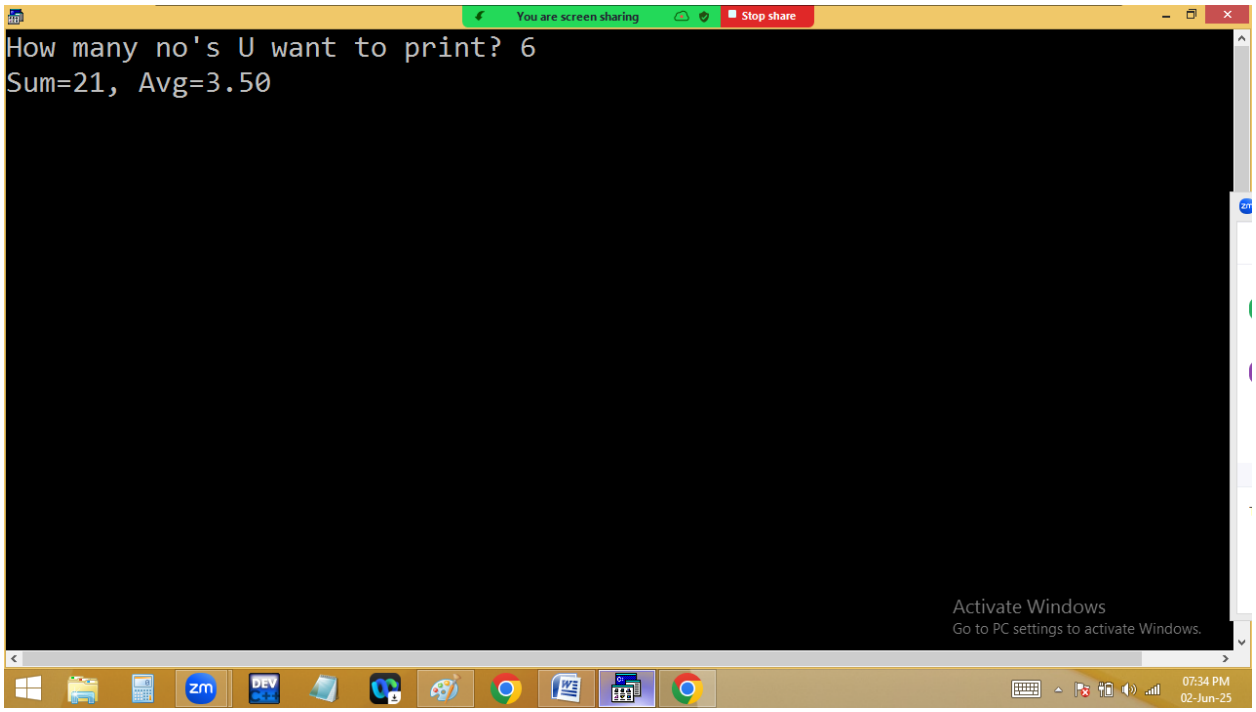
Without using loop?

S = n*(n+1)/2 ;

n=4 ➔ 4*(4+1)/2 ➔ 4*5/2=10




You are screen sharing
Stop share
How many no's U want to print? 4
Sum=10, Avg=2.50
Activate Windows
Go to PC settings to activate Windows.
07:34 PM
02-Jun-25


You are screen sharing
Stop share
How many no's U want to print? 6
Sum=21, Avg=3.50
Activate Windows
Go to PC settings to activate Windows.
07:34 PM
02-Jun-25

Home work:

1. **Finding power value using loop?**

Eg: 2^5=32

2. **Finding gcd / hcf of given two no's**

   Eg:

   4 factors are 1, 2, 4

   6 factors are 1, 2, 6

   2 is the gcd for 4 and 6